

ORF307

Least squares

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Today's topics

- Least-squares optimization
- Gram matrix
- Solving least squares problems
- Example

- **Recap**



Flop counts

- Flops → **F**loating-point **o**peration
- Computers store real numbers in floating-point format
- Basic arithmetic operations (+, −, ×, ÷) are called floating-point operations
- **Algorithm complexity** = **total computational cost** = **total number of flops needed as a function of dimensions** (of vectors, metrics, etc.)
- Modern computers can perform: **1 Gflops/sec** (10^9 flops/sec)

Summary of easy linear systems $Ax = b$

- **Diagonal:** $A = \text{diag}(a_1, a_2, \dots, a_n)$

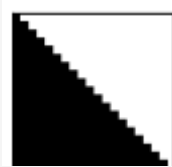
- Method: $x_i = b_i/a_i$



- Flops: n

- **Lower triangular:** $A_{ij} = 0, \forall j > i$

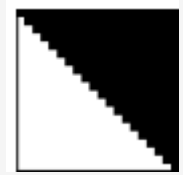
- Method: fwd substitution



- Flops: n^2

- **Upper triangular:** $A_{ij} = 0, \forall j < i$

- Method: bwd substitution



- Flops: n^2

- **Permutation:** $P_{ij} = 1, \text{ if } j = \pi_i; \text{ else } 0$

- Method: rearrangements

- Flops: 0

Multiple right-hand sides

- Many times, we need to solve d linear systems with the **same coefficient matrix** $A \in R^{n \times n}$ but **different right-hand sides** ($b_1, b_2, \dots, b_d \in R^n$)

$$Ax = b_1, \quad Ax = b_2, \quad \dots, \quad Ax = b_d$$

- Factorize** A **once**
- Solve** all linear systems using the **same (cached) factorization**
 - The 'solve' can be done in **parallel**.

LU factorization

- Every invertible matrix A can be factored as

$$A = PLU \quad \rightarrow \quad P^T A = LU$$

- P : **permutation**, L : **lower triangular**, U : **upper triangular**.

- **Procedure:**

- Similar to Gaussian elimination
- Permutation P avoids divisions by 0
- There are many combinations of P, L, U

- **Complexity:** $(2/3)n^3$ flops or $O(n^3)$

NOTE: Fewer flops if matrix A has **special structure** (e.g., lower/upper triangular or diagonal matrix)

LU solution

- Once the factorization $A = PLU$ has been computed, we can solve the system:

$$Ax = b \Rightarrow PLUx = b \Rightarrow x = U^{-1}L^{-1}P^{-1}b$$

- STEP #1.** permutation: $Py_1 = b \Rightarrow y_1 = P^{-1}b$ (0 flops)
- STEP #2.** fwd subs: $Ly_2 = y_1 \Rightarrow y_2 = L^{-1}y_1$ (n^2 flops)
- STEP #3.** bwd subs: $Ux = y_2 \Rightarrow x = U^{-1}y_2$ (n^2 flops)
- Complexity:**
 - Factor+solve** = $(2/3)n^3 + 2n^2 \approx (2/3)n^3$

Cholesky (LL^T) factorization

- If $A \succcurlyeq 0$ is **PSD matrix** it can be **factorized** as

$$A = LL^T$$

- L is a **lower triangular** matrix
- **NOTE #1:** works only when A is symmetric PSD matrix
- **NOTE #2:** there is no need for permutation
- **Complexity:**
 - $(1/3)n^3$ flops (half of LU factorization)

Cholesky (LL^T) solution

- Once the LL^T factorization has been computed we can solve the system

$$\Rightarrow Ax = b \Rightarrow LL^T x = b \Rightarrow x = L^{-T} L^{-1} b$$

- Iterations:**

- STEP* #1. fwd subs: $Ly_1 = b \Rightarrow y_1 = L^{-1}b$ (n^2 flops)
- STEP* #2. bwd subs: $L^T x = y_1 \Rightarrow x = L^{-T}b$ (n^2 flops)

- Complexity:**

- Factor+Solve:** $(1/3)n^3 + 2n^2 \approx (1/3)n^3$

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- **Least squares optimization**

❖ **Solution of systems of linear equations *and the Null Space***

The Null Space and Linear Independence

- **Definition:** The **null space** of a matrix $A \in R^{m \times n}$ is the set of vectors $x \in R^n$ that are mapped to the zero vector when multiplied by A

$$N(A) = \{x \in R^n \mid Ax = 0\}$$

The Null Space and Linear Independence

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- **Visualization:** Think of $N(A)$ as the set of all those “*poor vectors*” that when they meet A and are multiplied by it, they are smashed to nothing (zero).
- If the columns of A are **linearly independent** ($\text{rank}(A) = n$), then $N(A) = \{0\}$
- If the columns of A are **linear dependent**, then $N(A)$ **contains more than the zero vector**.

Case 1. Unique solution

- The system $Ax = b$ has a unique solution **iff**
 - The rhs b can be written as a linear combination of the columns of A
 - The null space $N(A) = \{0\}$ (**columns** of A are **linear independent**)
- **Example:** $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$, $b = \begin{pmatrix} 3 \\ 7 \\ 11 \end{pmatrix}$, $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Case 2. Infinitely many solutions

- The system $Ax = b$ has infinitely many solutions iff
 - The rhs b can be written as a linear combination of the columns of A
 - The null space $N(A)$ contains more than the zero vector.
- **Example:** $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$, $b = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $x = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + c \begin{pmatrix} -2 \\ 1 \end{pmatrix}$, where $c \in \mathbb{R}$
- **Note:** how many equations do you actually have in the above example?

Case 3: No solution

- The system $Ax = b$ has no solution iff
 - The rhs b **cannot** be written as a linear combination of the columns of A
 - The null space of A is **irrelevant** in this case. The columns of A can be linear independent or dependent
- **Example:** $A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 3 & 4 \end{bmatrix}$, $b = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $\nexists x$
- **Question:** How can we find an **approximate solution** x^* of the system $Ax = b$?

Perhaps a vector that makes $Ax^* - b \approx 0$

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- **Least squares**

Solving overdetermined systems

- A system of linear equations is **overdetermined** when you have more equations than variables ($m > n$)

$$Ax = b, \quad A \in R^{m \times n}$$

- In general, overdetermined systems have **no solution**
- **Example:**

$$Ax = b \Leftrightarrow \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \rightarrow \text{no solution!}$$

Least squares problem

- Since there may not be a solution, the **residual** $r = Ax - b \neq 0, \forall x \in R^n$
- **Goal:** Find an $x \in R^n$ that **minimizes some norm** of the residual:

$$\min ||r|| = \min ||Ax - b||$$

- **Least squares optimization problem** (use Euclidean (ℓ_2) norm):

$$\min ||Ax - b||_2^2$$

- Decision variables: $x \in R^n$ (i.e., the approximate solution of the system)
- Objective function: $f(x) = ||Ax - b||_2^2$

Least squares solution

$$\min f(x) = \|Ax - b\|_2^2$$

- **Optimality condition:** a point $x^* \in R^n$ is a **solution** of the least squares optimization problem iff

$$\|Ax^* - b\|_2^2 \leq \|Ax - b\|_2^2, \quad \forall x \in R^n$$

- **Remark:** the optimal solution x^* need not satisfy $Ax^* - b = 0$
 - **But** if $Ax^* - b = 0$, then x^* is a **solution** of the linear system $Ax = b$

Column interpretation

- We can write $A \in R^{m \times n}$ in column format: $A = [a_1, a_2, \dots, a_n]$, with $a_i \in R^m$
- Then: $\|Ax - b\|^2 = 0 \Leftrightarrow \|a_1x_1 + a_2x_2 + \dots + a_nx_n - b\|^2 = 0$
- **Interpretation:** find the **linear combination** (i.e., the x_i 's) of the **columns** of A that are **closest** to b .
- $x^* = (x_1^*, x_2^*, \dots, x_n^*)$ is the solution of the **optimization problem**

$$\min f(x) = \|Ax - b\|_2^2$$

- **Notation:** $x^* = \arg \min f(x)$

Row interpretation

- We can write $A \in R^{m \times n}$ in row format: $A = \begin{bmatrix} \tilde{a}_1^T \\ \vdots \\ \tilde{a}_m^T \end{bmatrix}$, where $\tilde{a}_i \in R^n$
- Then the residual vector $r = Ax - b \in R^m$ has **components**

$$r_i = \tilde{a}_i^T x - b_i, \quad i = 1, 2, \dots, m$$

- **Interpretation:** find a vector $x \in R^n$ that **minimizes the sum of squares**

$$\|Ax - b\|_2^2 = (\tilde{a}_1^T x - b_1)^2 + (\tilde{a}_2^T x - b_2)^2 + \dots + (\tilde{a}_m^T x - b_m)^2$$

- $x^* = (x_1^*, x_2^*, \dots, x_n^*)$ is the solution of the **optimization problem**

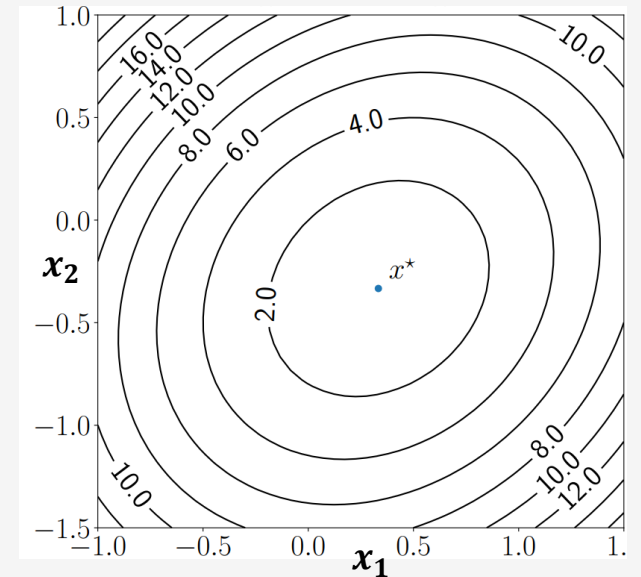
$$\min f(x) = \|Ax - b\|_2^2$$

Example

- Consider the linear system: $Ax = b \Leftrightarrow \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \rightarrow$ (no solution!)
- Solving the least squares optimization problem:

$$\min \|Ax - b\|^2 = (2x_1 - 1)^2 + (-x_1 + x_2)^2 + (2x_2 + 1)^2$$

we get the solution: $x^* = (1/3, -1/3)^T$



- Interpretations:**

- $\|Ax^* - b\|^2 = 2/3$ which is the **smallest possible value of the residual**: $\|Ax - b\|^2$
- $Ax^* = (2/3, -2/3 - 2/3)^T$ is the **closest linear combination of columns of A to vector b**

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- **Gram matrices**

Gram matrix

- Consider any matrix $A \in R^{m \times n}$ whose column format is:

$$A = [a_1, a_2, \dots, a_n], \quad a_i \in R^m$$

- The **Gram matrix** of A is defined as

$$A^T A = \begin{bmatrix} a_1^T \\ a_2^T \\ \vdots \\ a_n^T \end{bmatrix} [a_1, a_2, \dots, a_n] = \begin{bmatrix} a_1^T a_1 & a_1^T a_2 & \cdots & a_1^T a_n \\ a_2^T a_1 & a_2^T a_2 & \cdots & a_2^T a_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n^T a_1 & a_n^T a_2 & \cdots & a_n^T a_n \end{bmatrix}$$

NOTE: Very useful for the solution of least squares optimization problems!

Invertibility of the Gram matrix

- **Theorem:** the matrix $A \in \mathbb{R}^{m \times n}$ has linear independent columns $\Leftrightarrow$ its Gram matrix $A^T A$ is invertible.

Positive (semi)definiteness of Gram matrices

- **Theorem:** the Gram matrix $A^T A$ is always PSD ($A^T A \succcurlyeq 0$)
- **Proof:** $\forall x \in R^n$ we have: $x^T A^T A x = (Ax)^T A x = ||Ax||^2 \geq 0$ (definition of norms)

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- **Theorem:** the Gram matrix $A^T A$ is PD ($A^T A \succ 0$) **iff** the columns of A are linear independent (or equivalently $N(A) = \{0\}$).
- **Proof:** if the columns of A are linear independent then

$$Ax \neq 0, \quad \forall x \in R^n, \quad \text{and } x \neq 0$$

- Then: $x^T A^T A x = (Ax)^T A x = ||Ax||^2 > 0, \quad \forall x \neq 0$ (definition of norms)

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- **Solving least squares problems**

Calculus-based derivation of the optimal solution

$$\min f(x) = ||Ax - b||^2$$

- **Assumption:** A has linear independent columns . Then $A^T A$ is invertible
- The objective function can be written as

$$f(x) = ||Ax - b||^2 = \sum_{i=1}^m (\mathbf{a}_i^T x - b_i)^2 = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j - b_i \right)^2$$

- It can also be written in the compact form (or matrix-vector form):

$$f(x) = ||Ax - b||^2 = (Ax - b)^T (Ax - b) = x^T A^T A x - 2(A^T b)^T x + b^T b$$

Calculus-based derivation of the optimal solution

$$f(x) = \|Ax - b\|^2 = (Ax - b)^T (Ax - b) = x^T A^T A x - 2(A^T b)^T x + b^T b$$

- The solution of the least squares problem satisfies the conditions:

First Order Necessary Condition (FONC) $\left\{ \begin{array}{l} \nabla f(x^*) = 0 \end{array} \right.$ $\Leftrightarrow \begin{pmatrix} \partial f(x^*)/\partial x_1 \\ \partial f(x^*)/\partial x_2 \\ \vdots \\ \partial f(x^*)/\partial x_n \end{pmatrix} = 0$ $\Leftrightarrow 2A^T A x^* - 2A^T b = 0 \Leftrightarrow$

$\Leftrightarrow \underbrace{A^T A x^* = A^T b}_{\substack{n \times n \text{ system of} \\ \text{linear equations}}} \leftarrow \text{Normal Equations}$

Gradient of f

Remark: It is a Generalization of the **univariate function** case: $g'(x) = 0$ where $g: R \rightarrow R$

*Check for optimality of x^**

- **CLAIM:** $\|Ax^* - b\|^2 \leq \|Ax - b\|^2, \forall x \in R^n$ (i.e., x^* is the **minimum**!)

Check for optimality of x^*

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$$\|Ax - b\|^2 =$$

Check for optimality of x^*

- **CLAIM:** $\|Ax^* - b\|^2 \leq \|Ax - b\|^2, \forall x \in R^n$ (i.e., x^* is the **minimum**!)

$$\|Ax - b\|^2 = \|(Ax - Ax^*) + Ax^* - b\|^2 =$$

Check for optimality of x^*

- **CLAIM:** $\|Ax^* - b\|^2 \leq \|Ax - b\|^2, \forall x \in R^n$ (i.e., x^* is the **minimum!**)

$$\|Ax - b\|^2 = \|(Ax - Ax^*) + Ax^* - b\|^2 =$$

$$\|Ax - Ax^*\|^2 + \|Ax^* - b\|^2 + 2(Ax - Ax^*)^T(Ax^* - b) =$$

$$\|Ax - Ax^*\|^2 + \|Ax^* - b\|^2 + 2(x - x^*)^T \underbrace{A^T(Ax^* - b)}_{(= A^T Ax^* - A^T b = 0)} =$$

$$\|Ax - Ax^*\|^2 + \|Ax^* - b\|^2$$

Since we know: $\|Ax - Ax^*\|^2 \geq 0$ we can conclude:

$$\|Ax - b\|^2 \geq \|Ax^* - b\|^2$$



Solving the normal equations

- Normal equations: $A^T A x^* = A^T b$ Always check for linear independence of the columns of matrix A
- **Approach #1**: By **inverting** the matrix $A^T A$ we obtain: $x^* = (A^T A)^{-1} A^T b$
 - The matrix $(A^T A)^{-1} A^T$ is called pseudoinverse or Moore-Penrose of the matrix A
 - The matrix $(A^T A)^{-1}$ **exists** because A has linear independent columns
- **Approach #2**: use **Cholesky factorization** to get $A^T A = L L^T$
 - Then perform **bwd and fwd substitutions** to find x^*
- **Remark**: **Approach #2** is more computationally stable and efficient and used in practice!

Solving normal equations with Cholesky

- **Main steps of Approach #2 and their computational costs:**

1. Form the linear system: $A^T A x = A^T b$

- 1.1: form $M = A^T A$ ($2mn^2$ flops)

- 1.2: form $q = A^T b$ ($2mn$ flops)

2. Factor $M = LL^T$ ($(1/3)n^3$ flops)

3. Solve the linear system: $LL^T x = q$ ($2n^2$ flops)

(fwd/bwd subs) $\rightarrow$ (first perform fwd sub: $Ly = b$ and then bwd sub: $L^T x = y$)

Solving normal equations with Cholesky

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Complexity:

Factor+Solve:

$$2mn^2 + 2mn + (1/3)n^3 + 2n^2 \approx 2mn^2$$

Prefactored for a new b :

$$2mn + 2mn^2 \approx 2mn$$

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- **Example: optimal advertising**

Optimal advertising (VMLS book, p.234)

- m **demographic groups** (e.g., 18-22 yo, 23-33 yo, 34-44 yo, 45-65 yo, etc.)
- n **advertising channels** (e.g., ABC, CBS, CNN, Fox, NBC, PBS, BBC, DW, France24, etc.)
- $A \in R^{m \times n}$ the **reach matrix** $\rightarrow$ contains the demographic reach of the channels
 - A_{ij} the **number of views** per \$ spent of demographic group i at the channel j
- $v^{des} \in R^n$ the vector of **desired views** for each demographic
- $s \in R^n$ is the **spending in \$** on the n advertising channels (**not known**)
- $v = As \in R^m$ the **views** across the demographic groups (**non known**)
- **Goal:** Find s such that the squared deviation $\|As - v^{des}\|^2$ from the desired views is **minimized**

Optimal advertising *(VMLS book, p.234)*

Check GitHub companion for Python code

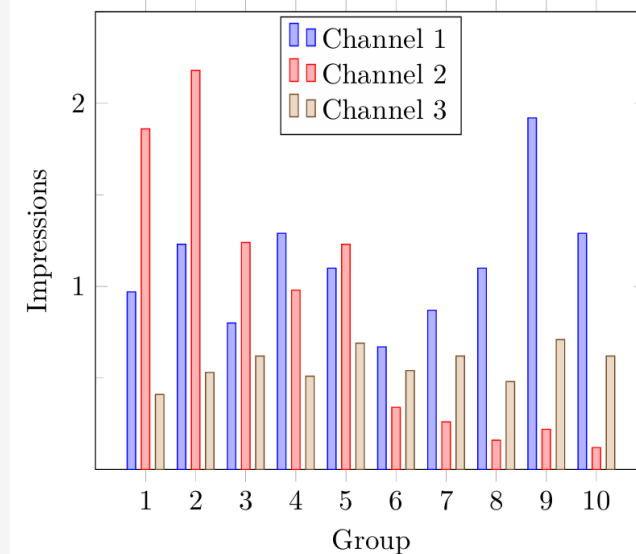
- Some representative data

- $m = 10, n = 3, v^{des} = (1000, 1000, \dots, 1000)^T \in R^{10}$

Channels

Groups	$A =$	0.97	1.86	0.41
		1.23	2.18	0.53
		0.80	1.24	0.62
		1.29	0.98	0.51
		1.10	1.23	0.69
		0.67	0.34	0.54
		0.87	0.26	0.62
		1.10	0.16	0.48
		1.92	0.22	0.71
		1.29	0.12	0.62

1K views / \$1



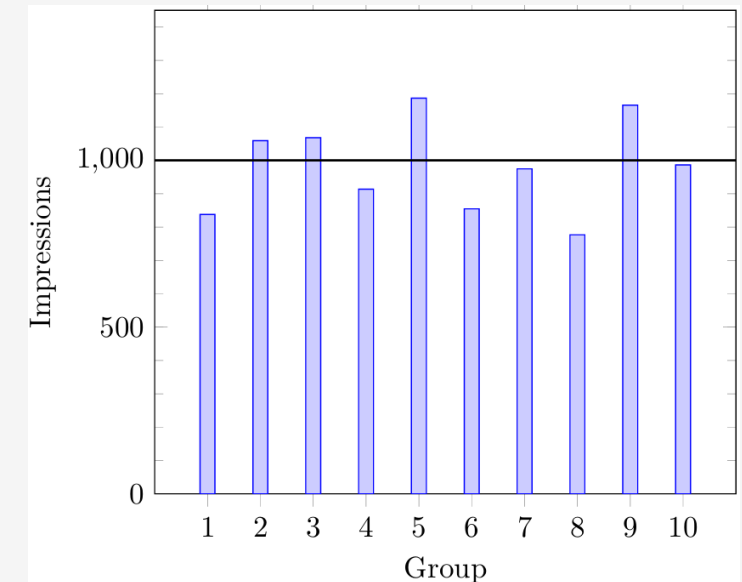
1K views / \$1

- Least squares problem

$$\min \|As - v^{des}\|^2$$

- $s^* = (62, 100, 1443)^T \in R^3$

Solving the Normal equations



- **Reusing the Cholesky factorization** LL^T for different v^{des}
- Suppose you have: $m = 100,000$ groups and $n = 5,000$ channels

**Pseudoinverse
Computation**

Time: 263 sec

First solve with $v_1^{des} = (10^3)\mathbf{1}$

1. Form linear system: $Ms = q$
Where: $M = A^T A$, $q_1 = A^T v_1^{des}$
2. Factor $M = LL^T$
3. Solve $LL^T s = q_1$

Complexity: $2mn^2$

Time: 9 sec

Second solve with $v_2^{des} = (500)\mathbf{1}$

1. Form $q_2 = A^T v_2^{des}$
2. Solve $LL^T s = q_2$

Complexity: $2mn$

Time: 0.37 sec

References

- S. Boyd, L. Vandenberghe: Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares
 - Ch. 12: Least squares

Next lecture

- Least squares and data fitting
